

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #2 Solutions

1. Let P = Pilgrim, A = Ant, F = Fly, and G = “Geo drops.” We are told that:

$$P(P) = 0.40, \quad P(A) = 0.35, \quad P(F) = 0.25,$$

$$P(G | P) = 0.6, \quad P(G | A) = 0.3, \quad P(G | F) = 0.8.$$

- (a) By the **Law of Total Probability**,

$$\begin{aligned} P(G) &= P(G | P)P(P) + P(G | A)P(A) + P(G | F)P(F) \\ &= 0.6(0.40) + 0.3(0.35) + 0.8(0.25) \\ &= 0.24 + 0.105 + 0.20 \\ &= 0.545 \end{aligned}$$

- (b) By **Bayes’ rule**,

$$P(A | G) = \frac{P(G | A)P(A)}{P(G)} = \frac{0.3 \cdot 0.35}{0.545} = \frac{0.105}{0.545} \approx 0.19266.$$

- (c) We are comparing the **prior** $P(F)$ to the **posterior** $P(F | G)$.

$$P(F) = 0.25, \quad P(F | G) = \frac{P(G | F)P(F)}{P(G)} = \frac{0.8 \cdot 0.25}{0.545} = \frac{0.20}{0.545} \approx 0.36697.$$

Since $0.36697 > 0.25$, knowing Geo dropped makes it **more likely** the enemy was a Fly.

2. Jason and Stevie are independent and uniform on $\{0, 1, 2, 3, 4, 5\}$. Let their scores be J and S . We can make a table of J and S and the corresponding total:

$J \setminus S$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	6
2	2	3	4	5	6	7
3	3	4	5	6	7	8
4	4	5	6	7	8	9
5	5	6	7	8	9	10

- (a) Since each outcome in the table is equally likely, we can compute the conditional probabilities $P(A | B) = \frac{P(A \cap B)}{P(B)}$ just by counting the numbers of items in each set. For example, here there are 18 outcomes where S is odd (blue text), 6 outcomes where the sum is greater than equal to 8 (bold text), and 4 outcomes where both conditions are true. Thus,

$$\begin{aligned} P(J + S \geq 8 \mid S \text{ is odd}) &= \frac{P(J + S \geq 8 \text{ and } S \text{ is odd})}{P(S \text{ is odd})} \\ &= \frac{\frac{4}{36}}{\frac{18}{36}} \\ &= \frac{4}{18} \\ &= \frac{2}{9} \end{aligned}$$

- (b) To show that A and B are independent, we must show that $P(A \cap B) = P(A) P(B)$. Again, by counting outcomes in the table, we can compute that $P(J + S \text{ is odd}) = \frac{1}{2}$, $P(J \text{ is odd}) = \frac{1}{2}$, and $P(J + S \text{ is odd} \cap J \text{ is odd}) = \frac{1}{4}$, so the two events are independent.
- (c) We can compute that $P(J = 2) = \frac{1}{6}$, $P(J + S \geq 5) = \frac{7}{12}$, and $P(J = 2 \cap J + S \geq 5) = \frac{1}{12}$. Since $\frac{1}{6} \cdot \frac{7}{12} = \frac{7}{72} \neq \frac{1}{12}$, the two events are not independent.
3. Each shop independently contains at least one Rare Joker with probability $p = 0.4$. Let $q = 1 - p = 0.6$.
- (a) There are $\binom{8}{3} = 56$ ways for exactly 3 of the 8 shops to contain Rare Jokers, and each of these events has probability $\left(\frac{4}{10}\right)^3 \left(\frac{6}{10}\right)^5$, so the overall probability is $\binom{8}{3} (0.4)^3 (0.6)^5 = 56(0.064)(0.07776) \approx 0.279$.
- (b) If Y is the number of Rare Jokers, then

$$\begin{aligned}
 P(Y \geq 3) &= 1 - (P(Y = 0) + P(Y = 1) + P(Y = 2)) \\
 &= 1 - \left[\binom{6}{0} q^6 + \binom{6}{1} p q^5 + \binom{6}{2} p^2 q^4 \right] \\
 &= 1 - [(0.6)^6 + 6(0.4)(0.6)^5 + 15(0.4)^2(0.6)^4] \\
 &\approx 1 - (0.046656 + 0.186624 + 0.311040) \\
 &\approx 0.456
 \end{aligned}$$

- (c) The first success occurring at the 5th shop means that we failed 4 times and then succeeded, i.e.,

$$P(\text{first Rare Joker at 5}) = q^4 p = (0.6)^4 (0.4) = 0.1296(0.4) = 0.052.$$

- (d) Bernoulli's theorem says that

$$P\left(\left|\frac{k}{n} - p\right| > \epsilon\right) < \frac{p(1-p)}{n\epsilon^2}$$

or equivalently

$$P\left(\left|\frac{k}{n} - p\right| < \epsilon\right) > 1 - \frac{p(1-p)}{n\epsilon^2}$$

Applying this to our problem, $p = \frac{4}{10}$ and $n = 12000$. We expect to get $12000 \cdot \frac{4}{10} = 4800$ wins, so the condition $k \in [4600, 5000]$ is equivalent to $\left|\frac{k}{n} - p\right| < \frac{200}{12000}$, making $\epsilon = \frac{1}{60}$. So the lower bound from the right hand side is

$$1 - \frac{\frac{4}{10} \cdot \frac{6}{10}}{12000 \left(\frac{1}{60}\right)^2} = 0.928$$

So this is quite likely according to Bernoulli's Theorem. (Later in the semester we'll see how we can get an even more accurate value for this probability).